

DEMO DAY · 25 MINUTES

Tie Me Ghana Speaker Notes

The opening words. What to say on each screen. And how to answer the question, "Where can we use this?"

How to use these notes

Every screen in this pack is laid out the same way. Once you learn the shape, you can find your place on any page in one second.

YOU WILL SEE	IT MEANS
On this screen	What this screen is, and the one point it proves. Know this before you talk.
They see	What is in front of the judges right now.
Say this	The green box. These are the words. You can read them as they are.
Then	The tap you make next.
Move on with	The one line that carries you into the next screen. Do not skip it — this is what makes the demo feel like one story and not ten stops.

- **Speak slowly.** Slow sounds sure. Fast sounds nervous.
- **Look at the judges, not the screen.** You know the app. They do not.
- **Say what a screen is for before you show it.** One line is enough.
- **Never tap in silence.** Talk while the screen loads.
- **Do not say sorry** if something is slow. Just keep talking.

Before you go on stage

- Open the site on both devices and let the sign videos play once. They are then saved on the device and load fast.
- Put the phone and the laptop on the same Wi-Fi.
- Turn the sound up. One part of the demo speaks out loud.
- Have the demo already open at screen 1 before you say a word.
- Make sure the newest build is live on the web site.
- Agree who speaks and who taps. One voice, one pair of hands.

Questions that play as one filmed clip

Type one of these and the patient sees a single, smooth sign video of the whole question. Anything else is built word by word, which still works but does not look as good. Stay on this list on stage.

- Where does it hurt?
- Where is your pain?
- Do you have fever?
- Are you vomiting?
- Do you feel dizzy?
- Do you have allergies?
- Do you have a similar problem before?

Six of these are new this week. If a judge asks how hard it is to add a sign, that is your answer: we filmed them and imported them. No code changed.

The shape of the talk

Read this page once and you will not get lost. The demo is one story in three parts: **the patient who cannot read**, then **the patient who can**, then **what the hospital has to buy** — which is nothing.

1	Does she have a phone?	We do not punish the poorer patient	40s
2	The pairing code	Two devices, and no server in the middle	50s
3	Can she read?	The one question that changes everything after it	50s
4	She cannot read	Signs, a body map, a pain scale — no words at all	2m
5	She can read	Free sentences, and the doctor keeps his eyes up	70s
6	The app says no	We refuse a half sentence. This is the safety story	60s
7	Emergency	One tap, both screens, before any form	60s
8	The medicine	She takes it home on her own phone. The strongest moment	2m
9	One phone, shared	The same visit for a patient with no phone	40s
10	No app store	A hospital needs a link, not new machines	40s

That is about ten minutes with nothing rushed. If you are running late, cut screen 5 and screen 9. Never cut 6 or 8.

The opening (about 90 seconds)

STAND STILL. DO NOT TOUCH THE LAPTOP YET.

Good morning. We are team **[team name]**. Our project is called **Tie Me Ghana**.

Think of a deaf patient who walks into a hospital in Ghana today.

The doctor speaks. She cannot hear him. So he writes it down. But many deaf people in Ghana did not finish school, so she cannot read it either.

Then the hospital looks for a sign language helper. There are very few. Most days there is none. So the visit goes on with hand waving and guessing.

That is the problem. A patient who cannot say where the pain is. A doctor who cannot be sure she understood the medicine.

Tie Me Ghana lets those two people talk. Today. With the phone that is already in their hand.

MOVE ON WITH

"Let me take you through one visit, from the door to the pharmacy."

Three lines that add weight

Use one of these in the opening if you want the problem to land harder. Only say a number you have checked yourself.

- "Ghana has about **[number]** deaf people, and only **[number]** trained sign language helpers."
- "A hospital cannot hire a helper for every ward. But every ward already has a phone."
- "This is not a new gadget. It is the gap between a doctor and a patient, closed with what the hospital already owns."

Part one: the patient who cannot read

1 “Does this patient have their own phone?”

40 sec

On this screen: the first question of every visit. It is here to show that we thought about the patient who owns nothing.

They see: two big buttons. No form, no name, no login.

SAY THIS

This is the first thing the app asks the nurse. Not the patient’s name. Not a form. Just this.

If she has a phone, we use two screens. The doctor keeps his. She holds her own.

If she has no phone, we use one screen and turn it between them.

Both ways are the full app. We did not want the poorer patient to get the weaker one.

Then: tap *Yes, pair it*.

MOVE ON WITH

“This patient has a phone. So let us join the two together.”

2 The pairing code

50 sec

On this screen: the two devices join. It is here to tell the privacy story early, before anyone has to ask for it.

They see: six letters on the doctor’s screen. You type them into the phone. Then “Connected”.

SAY THIS

Six letters. She types them once, and the two devices are joined.

Now the important part. From here the two devices talk **straight to each other**.

Nothing the doctor asks and nothing the patient answers goes through our server.

Our server only helped them find each other, the way you give someone a phone number. Once the call starts, it hangs up.

So the questions, the body map, the medicines — we never hold any of it. In a hospital that is not a clever choice. It is the only one.

Then: type the code on the phone. Wait for “Connected” to show on both.

MOVE ON WITH

“They are joined. Now the app asks the one question that decides everything after it.”

On this screen: the fork in the road. It is here to show that we ask the patient, instead of letting staff guess.

They see: the same question on both devices at the same moment — and on her phone it is signed, not written.

SAY THIS

Here is the question everything else hangs on.

Look at her phone. The question is there too, in sign language. Because this is a question **about her**. She should answer it herself, not have a nurse guess for her.

If she reads, we give her words. If she does not, we give her signs, a body map and pictures. Same app. Two roads.

Then: tap *No* on the phone. Always show the harder road first — that is the one the judges came to see.

MOVE ON WITH

"She does not read. So we take the road with no words on it at all."

On this screen: the heart of the demo. It is here to prove a doctor can take a history from a patient who cannot read one word.

They see: the doctor taps a question. A real person signs it on her phone. She taps *Yes* or *No* — and those two buttons carry their own sign clips. Then a body map, then a pain scale.

SAY THIS

The doctor asks, “Do you have pain?” He taps it. He does not type it.

On her phone, a real person signs the question. That is Ghanaian Sign Language. Not American. We filmed it here.

Now look at the two buttons. Even *Yes* and *No* are signed. A patient who reads nothing at all can still answer this screen.

She taps. Her answer lands on the doctor’s side in plain English.

Next, the body map. She points at where it hurts, on a drawing of a body. Then how bad it is, on a scale of faces.

In under a minute the doctor knows three things. There is pain. It is here. It is this bad. And not one word was written down.

Then: ask “*Do you have fever?*” — it plays as one clip. Then the body map, then the pain scale. Three things, no more. This is where teams run out of time.

MOVE ON WITH

“That is the patient who cannot read. Now the same doctor, with a patient who can.”

Part two: the patient who can read

5 Free words, and a voice for the doctor

70 sec

On this screen: the second road. It is here to show the app is not a fixed list of questions.

They see: the doctor types his own sentence. It comes out in Twi as text and as sign video. She types back, and the doctor's device speaks her words out loud.

SAY THIS

This road is for a patient who reads. Here the doctor is not stuck with our questions. He types what he wants to ask.

We show it two ways at once. In Twi, as text. And as sign video, for the words we have filmed.

Now listen. When she types back, the doctor's device **says it out loud**.

That is on purpose. It means the doctor can keep his eyes on his patient instead of on a screen. Which is what a doctor is supposed to do.

Then: tap *New patient*, choose *Yes*, *pair it*, and answer *Yes* to the reading question. Type "*Has your pain hurt since the morning?*" — seven signs, and nothing in it can refuse. Sound up.

MOVE ON WITH

"Now watch what happens when the doctor asks for something we cannot sign properly."

On this screen: our safety rule. It is here because it is the part that separates a health tool from a demo. Slow down.

They see: a sentence with a word we have not filmed. The app **will not play it**, and says which word is missing.

SAY THIS

This is the part we are most proud of. And it is the part that does nothing at all.

I have typed, "your pain is not serious". The word "not" has no sign in our library yet.

A normal app would drop that word and play the rest. The patient would see "your pain is serious". The opposite of what I said.

In a hospital, that is how somebody gets the wrong dose.

So we refuse. The app tells the doctor which word is missing and asks him to say it another way.

We would rather show nothing than show half of a sentence.

Then: type "*Your pain is not serious.*" It refuses on "not". Pause after the last line and let it sit. The full list of sentences is at the back of these notes.

MOVE ON WITH

"That is a normal visit. Now let me show you a bad night."

On this screen: the worst case. It is here to show the app works when there is no time for any of the above.

They see: the doctor taps *Emergency* and both screens change in the same moment. Four big red buttons.

SAY THIS

Now picture a deaf patient who arrives at night. Alone. And cannot breathe.

One tap, and both screens change together. Nobody has to explain it to her. Her phone is already showing it.

Four buttons. Big. Red. She taps one, and the nurse has it in English straight away.

And this screen is open before the visit even begins. You should not have to finish a form to say you cannot breathe.

Then: tap one red button on the phone so the judges watch it land on the doctor's side.

MOVE ON WITH

"The visit is over. The doctor knows what is wrong. Now the part everyone forgets."

Part three: what she takes home

8

The medicine, on her own phone

2 min

On this screen: the strongest moment you have. It is here because this is the problem that does not stop at the hospital gate.

They see: the doctor builds a short medicine list. On her phone: sign videos for each one, a *Save* button, and a QR code.

SAY THIS

This is where most visits fail, for everybody, deaf or not. The patient walks out, and by the gate she has forgotten half of it.

The doctor picks the medicine, how much, and how often. Three taps.

Now look at her phone. She gets it as sign video. One clip for each instruction. She can play it as many times as she likes.

And she taps *Save*. It is on her phone now. She can watch it again tonight, at home, when the doctor is not there.

The QR code is for the pharmacy, so nobody has to read a doctor's handwriting.

That is the whole visit. She arrived unable to say where it hurt. She leaves knowing what to take.

Then: tap *Save* and show the file is really there. Do not rush this one.

MOVE ON WITH

"That was a visit with two phones. Most patients will have one screen between them. So here is the same visit again."

9 One phone, shared

40 sec

On this screen: the answer to “what if she has no phone?” before a judge asks it.

They see: one screen split by a line. “Doctor” on one side, “Patient” on the other.

SAY THIS

Many patients will not have a smart phone. So here is the same visit on one screen.

See the line down the middle. This half is the doctor’s. That half is hers. Turn the screen, and each person knows which part is theirs.

A small thing. We added it because we watched people get confused without it.

Then: keep it short. Show the line and move on.

MOVE ON WITH

“One last question you are all thinking. What does a hospital have to buy to run this?”

10 No app store. No download.

40 sec

On this screen: the close. It is here to leave them with one number — nothing.

They see: *Add to Home Screen* in the browser, and then the app icon on the phone.

SAY THIS

Nothing. There is nothing to install.

It is a web site. It opens in any browser, on any phone, on whatever old laptop the ward already has.

But add it to the home screen and it becomes an app with an icon. The sign videos are saved on the phone after the first time, so it stays fast even on a weak network.

A hospital does not need new machines for this. It needs the link.

MOVE ON WITH

“So that is Tie Me Ghana.”

The closing (30 seconds)

So that is Tie Me Ghana.

A deaf patient can say where the pain is. A doctor can be sure she understood the medicine. And nobody had to find a sign language helper.

It runs today, on the phones people already carry, in Ghanaian Sign Language we filmed here.

It is a pilot. It is not approved for clinical use yet, and we say so on every screen until it is.
Thank you. We are happy to take your questions.

Then: stop talking. Do not add anything. Let the silence sit until the first question comes.

“Where can we access it?”

Expect this one. Answer it fast and clearly. Say the link out loud, letter by letter if you must, and put it on the screen while you say it.

Say this

It is live right now. You can open it on your own phone, in this room.

The doctor’s side is at **tiemeghana.netlify.app**.

The patient joins at **tiemeghana.netlify.app slash join**.

There is nothing to download. No app store. It is a web site, so it opens in Chrome, in Safari, in any browser, on a phone, a tablet or a laptop.

And if you want it to look like an app, open it once and tap “Add to Home Screen”.

WHO	WHERE THEY GO
Doctor or nurse	https://tiemeghana.netlify.app
Patient	https://tiemeghana.netlify.app/join
A judge who wants to try it now	Open the first link and choose “No, share this device”. You do not need a second phone.

The harder versions of that same question

Q. Is this a real deployment, or a demo you run on your laptop?

"It is a real deployment. The site is hosted online, the database is online, and the sign videos come from cloud storage. What you are watching is not running on this laptop."

Q. Can we see the code?

"Yes. The code is in our repository and we are happy to give the judges access. Tell us which email to add." Do not promise a public link on stage if the repository is private.

Q. How do we reach your team after today?

"I am [name], and the best way is [email or phone]. We answer the same day." Agree on one person and one address before you go on stage. Never work it out in front of the judges.

Q. Is it in an app store?

"No, and that is on purpose. An app store means a download, a data bundle, and a phone with space on it. A link works on every phone today. We can still package it later if a hospital asks."

Other questions they will ask

Keep every answer to three lines. If you do not know, say "We have not solved that yet" and move on. Judges trust that more than a guess.

Q. How many signs do you have? Is that enough for a real hospital?

"We have 108 reviewed clips today, filmed with deaf signers. That is enough for the questions in this demo, and not enough for a whole hospital. That is exactly why the app refuses a sentence it cannot sign properly instead of guessing. Adding a sign means filming it and having it reviewed. It does not mean changing the code — we added forty-four more this week."

Q. Why not use a computer to make the signs?

"Sign language is face and body, not only hands. A machine avatar gets it wrong, and in a hospital wrong is dangerous. So every clip is a real deaf signer, and a reviewer has to approve it before any patient sees it."

Q. What about patient privacy? This is health data.

"Once the two devices are joined, they talk straight to each other. What the doctor asks and what the patient answers never reaches our server. We hold the joining code for a few minutes, and nothing else."

Q. Does it work without internet?

"Partly, and we will be straight with you. The sign videos are saved on the phone after the first use, so they play with no network. Joining two devices needs a network. We would rather say that than promise offline and fail on a ward."

Q. Did deaf people help you build this?

"The clips are signed by deaf signers, and a reviewer approves every clip before it goes live. The app will not show a clip that nobody approved."

Q. How does a hospital pay for this?

"A hospital pays far more for one sign language helper than for this. Our plan is a small yearly fee per hospital, with the clip library shared across all of them. Adding the tenth hospital costs us close to nothing."

Q. What is not finished?

"Three things. The clip library is small. Two devices need a network to join. And it is not approved for clinical use — it is a pilot, and we say so on the screen. We would rather list those than pretend."

Q. Can it do other Ghanaian languages?

"The text already goes to Twi. The sign videos are Ghanaian Sign Language, which is one language for the whole country, so that part does not change from region to region."

Q. What did you build, and what did you just use?

"We built the app, the sign matching, the safety rule, and the link between the two devices. We use an outside service to turn text into Twi. The sign clips are ours."

Small words instead of big ones

If you catch yourself saying the word on the left, say the one on the right. The judges will be tired by the tenth team. Plain words win.

DO NOT SAY	SAY
Peer-to-peer connection	The two devices talk straight to each other
Utilise, leverage	Use
Facilitate communication	Help them talk
Robust, seamless, cutting-edge	(say nothing — show it instead)
Deploy, deployment	Put it online, it is online
Our solution	Our app, or just "it"
Stakeholders	Doctors, nurses, patients
Scalable	Adding the next hospital costs us almost nothing
Validation, the gate	The app checks it, and refuses if a word is missing
Onboarding	Getting started

If something goes wrong on stage

- **A video is slow.** Keep talking. Say: “It is loading for the first time. After this it is saved on the phone.” Then go on.
- **The two devices will not join.** Say: “Let me show you the one-screen way, which is what most wards will use anyway,” and switch to the shared screen. Do not try the code three times.
- **A screen freezes.** Reload it. The app comes back to the same visit. Say: “It comes back to where we were — a nurse should not lose a visit because a phone went to sleep.”
- **You lose your place.** Go to screen 8, the medicine. It is the strongest part. Finish there.
- **Time runs out.** Stop the demo and say the closing lines. Never run past your time. Judges take marks for it.

Sentences you can type on the day

Every line below was run against the live clip library. The app has 108 signs, so it cannot say everything yet — and it will refuse rather than guess. Stay on this page and nothing will surprise you on stage.

The one trap to know

Type the filmed questions **word for word**. One small word out of place and the app builds the sentence by hand instead, or refuses.

THIS PLAYS ONE CLIP

Do you have fever?

Is **the** pain severe?

THIS REFUSES

Do you have **a** fever?

Is **your** pain severe?

“Fever” and “severe” are only filmed inside those questions, not as words of their own. So the shorter sentence works and the longer one does not. Read them off this page rather than from memory.

Best: these play as one smooth clip

One filmed person signs the whole question. This is what you want the judges to see.

TYPE THIS

Where does it hurt?

Where is your pain?

Do you have fever?

WHAT PLAYS

1 clip

1 clip

1 clip

Are you vomiting?	1 clip
Do you feel dizzy?	1 clip
Do you have allergies?	1 clip
Do you have any allergies?	1 clip
Do you have a similar problem before?	1 clip
Where exactly is the pain?	1 clip
When did the pain start?	1 clip
Is the pain severe?	1 clip
Have you had this problem before?	1 clip
Have you taken any medicine?	1 clip
Can you breathe normally?	1 clip

Also new: single words that have their own sign

Type just the word. One sign plays. Useful when a judge asks whether the app knows a symptom.

headache stomachache pain rashes itchy tired dizzy cold hot breathe
difficult tablet injection hospital insurance

A bonus you can show with these

Type **“Where is your pains?”** or **“Do you have fevers?”** — with the extra “s”. They still play the right clip.

Sign language does not add an “s” for more than one. So the app matches the word back to its base form. A doctor should not have to type carefully for a patient to be understood.

Good: these are built word by word

The app joins the signs one after another. It works, it just takes a few seconds longer. Use these to show the app is not a fixed list of questions.

TYPE THIS	WHAT PLAYS
No pain.	2 signs — shows a “no” is signed, not dropped
Where is the pain?	4 signs
When is the pain?	4 signs
How is your pain?	4 signs
Did the pain hurt?	4 signs

Your pain is over.	4 signs
The pain is below.	4 signs
No pain at night.	4 signs
My pain is at night.	5 signs
Since when is your pain?	5 signs
Your pain is in the morning.	6 signs
Is your pain in the morning?	6 signs
The pain is in the afternoon.	6 signs
Your pain is before the evening.	6 signs
Was the pain in the evening?	6 signs
Has your pain hurt since the morning?	7 signs
Did your pain hurt in the night?	7 signs
Are you tired?	3 signs
Are you hot?	3 signs
Are you dizzy?	3 signs
Can you breathe?	3 signs
Do you have pain?	4 signs
Are you in pain?	4 signs
Do you have a headache?	4 signs
Do you have a stomachache?	4 signs
Do you have a cold?	4 signs
Do you have rashes?	4 signs
Do you have hospital insurance?	4 signs
Do you have pain in the morning?	7 signs

For screen 6: these will refuse, on purpose

You need one of these for the safety part of the demo. Pick one, practise it, and use only that one.

TYPE THIS	WHY IT REFUSES	BEST FOR
Your pain is not serious.	"not" has no sign, and "serious" is not filmed	USE THIS The meaning flips if "not" is dropped

Do not take paracetamol.	"not", "take" and the drug name are all missing	The medicine story
Take two tablets at night.	"take", "two" and "tablets" are not filmed	Showing a dose cannot be half-signed

Watch what it does. It will not play this. It tells me the word "not" has no sign yet.

If it dropped that one word, the patient would read "your pain is serious". The opposite of what I said.

So it stops, and asks me to say it another way. That is the rule in this whole app: show nothing rather than show half.

Do not type these

They look safe and they are not. They will refuse in front of the judges.

DO NOT TYPE	WHY	TYPE THIS INSTEAD
Do you have a fever?	"fever" alone is not filmed, only the whole question	Do you have fever?
Is your pain severe?	"severe" alone is not filmed	Is the pain severe?
Did you take the tablet?	"take" has no sign yet	tablet
Do you need an injection?	"need" has no sign yet	injection
Is it itchy?	"it" has no sign yet	itchy
Are you pregnant?	"pregnant" is only filmed as the emergency button	Use the Emergency screen
Do you have malaria? / a cough?	no sign for either yet	Do you have fever?
Is the pain over your heart?	no body-part signs yet	Use the body map instead

Body parts are not filmed, so point at the body map for location. That is what it is there for.

If a judge types something of their own and it refuses

This will happen. It is a good moment, not a bad one. Do not apologise.

Good — that is the app doing its job. We have 108 signs today, so it will not say everything yet.

What matters is that it told you which word is missing, instead of playing a sentence that looks right and is not.

Adding that word is filming it, not rebuilding the app. We added forty-four this week.

One last thing

Do not read all of this on stage. Read the green boxes once the night before, and twice in the morning. On stage, look up.

The app works. You only have to get out of its way.